WORKED SOLUTION

STEP II 2014, Question 8

Pure • Greatest Binomial Coefficient

STEP II 2014, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q8

The key idea

To find where a sequence of positive terms peaks, compare each term with the next: look at the ratio c_{r+1}/c_r. For a binomial expansion this ratio is a simple fraction in r that decreases steadily, so the coefficients rise while it exceeds 1 and fall once it drops below 1. The largest coefficient sits where the ratio crosses 1, and every later part is a matter of reading off the integer part of one expression.

Solution

The coefficients and their ratio

By the binomial theorem,

$$c_r = \binom{n}{r} a^{n-r} b^r, \qquad 0 \le r \le n.$$

All of these are positive, since $a, b > 0$. For $0 \le r \le n-1$,

$$\frac{c_{r+1}}{c_r} = \frac{\binom{n}{r+1}}{\binom{n}{r}} \cdot \frac{b}{a} = \frac{(n-r)\,b}{(r+1)\,a},$$

using $\binom{n}{r+1} = \binom{n}{r}\frac{n-r}{r+1}$. Write

$$L = \frac{b(n+1)}{a+b}.$$

Because $(r+1)a > 0$,

$$\begin{aligned} c_{r+1} \ge c_r &\iff (n-r)b \ge (r+1)a \iff nb - a \ge r(a+b) \\ &\iff r \le \frac{nb-a}{a+b} = \frac{b(n+1)}{a+b} - 1 = L - 1, \end{aligned}$$

and the same chain holds with $=$ throughout. So

$$c_{r+1} > c_r \text{ if } r < L-1, \qquad c_{r+1} = c_r \text{ if } r = L-1, \qquad c_{r+1} < c_r \text{ if } r > L-1. \qquad (*)$$

The bounds on m

Upper bound. If $m = 0$ then $m \le L$ because $L > 0$. If $m \ge 1$, then $c_m \ge c_{m-1}$ because c_m is greatest, and by $(*)$ with $r = m-1$ this gives $m - 1 \le L - 1$, that is, $m \le L$.

Lower bound. If $m = n$ then $m \ge L - 1$ because $L < n + 1$ (as $\frac{b}{a+b} < 1$). If $m \le n - 1$, then $c_m \ge c_{m+1}$, and by $(*)$ with $r = m$ this forces $m \ge L - 1$.

So in every case

$$\frac{b(n+1)}{a+b} - 1 \le m \le \frac{b(n+1)}{a+b},$$

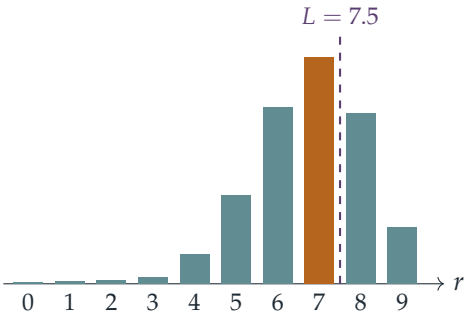
as required.

Ask yourself Why can't there be three values of m, or two that are not next to each other?

The interval $[L-1, L]$ has length 1. If L is not an integer, the interval contains exactly one integer, $\lfloor L \rfloor$, so m is unique. If L is an integer, it contains exactly the two consecutive integers $L - 1$ and L, so m is one of these. (Indeed $(*)$ with $r = L - 1$ shows $c_{L-1} = c_L$, so both really are greatest.) Either way, the unique value or the larger of the two is

$$G(n,a,b) = \left\lfloor \frac{b(n+1)}{a+b} \right\rfloor.$$

This formula is all we need from now on.



The coefficients of $(1+3x)^9$, to scale: they rise while $r < L - 1 = 6.5$ and then fall, peaking at $r = 7$.

Part (i)

For $(n,a,b) = (9,1,3)$, $L = \frac{3\times 10}{4} = 7.5$, so $G(9,1,3) = 7$.

For $(n,a,b) = (9,2,3)$, $L = \frac{3\times 10}{5} = 6$, an integer, so the greatest coefficient occurs at both $r = 5$ and $r = 6$, and $G(9,2,3) = 6$. (Check: $c_5 = 126 \cdot 2^4 \cdot 3^5 = 489\,888 = 84 \cdot 2^3 \cdot 3^6 = c_6$.)

Part (ii)

With $a = b$, $L = \frac{n+1}{2}$.

- $n = 2k$: $L = k + \frac{1}{2}$, so $G(2k, a, a) = k$ (the middle coefficient, unique).
- $n = 2k - 1$: $L = k$, an integer, so the two middle coefficients $r = k - 1$ and $r = k$ tie, and $G(2k - 1, a, a) = k$.

This matches the symmetry of the rows of Pascal's triangle.

Part (iii)

For fixed n and b, $L = \frac{b(n+1)}{a+b}$ is a decreasing function of a, so $\lfloor L \rfloor$ is a non-increasing function of a. Over the positive integers it is therefore greatest at $a = 1$, where $G(n, 1, b) = \left\lfloor \frac{b(n+1)}{b+1} \right\rfloor$. (Other values of a may tie with it, but none does better.)

Part (iv)

Now

$$\frac{b(n+1)}{b+1} = (n+1) - \frac{n+1}{b+1}.$$

Since $\frac{n+1}{b+1} > 0$, this is less than $n + 1$, so $G(n, 1, b) \leq n$. (This also follows from $m \leq n$.) Moreover,

$$G(n, 1, b) = n \iff (n+1) - \frac{n+1}{b+1} \geq n \iff \frac{n+1}{b+1} \leq 1 \iff b \geq n.$$

So the greatest possible value of $G(n, 1, b)$ is n, and it is achieved exactly when $b \geq n$. When $b = n$ the last two coefficients tie, since $c_{n-1} = n \cdot b^{n-1} = b^n = c_n$; for $b > n$ the last coefficient b^n is the unique largest.

Checks

- The coefficients of $(1 + 3x)^9$ are $1, 27, 324, 2268, 10206, 30618, 61236, 78732, 59049, 19683$: the largest is at $r = 7$, as found in (i).
- For $(1 + x)^4$ ($n = 2k$, $k = 2$) the coefficients $1, 4, 6, 4, 1$ peak at $r = 2 = k$; for $(1 + x)^3$ ($k = 2$) they are $1, 3, 3, 1$, with the tie at $r = 1, 2$ and larger value $2 = k$.
- In (iv) with $n = 3$: $(1 + 2x)^3$ has coefficients $1, 6, 12, 8$, peak at $r = 2$; $(1 + 3x)^3$ has $1, 9, 27, 27$, so $G = 3$ is reached once $b = 3$.
- A computer check of $G(n, a, b) = \lfloor b(n + 1)/(a + b) \rfloor$ over all $n < 30$ and $a, b < 15$ agreed in every case.

Common slips

- Forgetting the end cases $m = 0$ and $m = n$ when proving the bounds: there is no c_{-1} or c_{n+1} to compare with.
- Taking the smaller of the two tied values when L is an integer: G is defined as the larger, so $G(9, 2, 3) = 6$, not 5.
- In (iv), answering only $b = n$: every $b \geq n$ gives $G = n$.

Beyond STEP

The ratio test used here, comparing consecutive terms, is the same idea as the ratio test for convergence of series, and it locates the mode of the binomial distribution $\mathrm{B}(n, p)$ with $p = \frac{b}{a+b}$: the mode is $\lfloor (n+1)p \rfloor$. Sequences like the binomial coefficients, whose terms rise and then fall, are called unimodal, and log-concavity ($c_r^2 \geq c_{r-1}c_{r+1}$) is the usual tool for proving this. For the probability side, see Grimmett and Welsh, *Probability: An Introduction*.

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